

Qiyao Peng, ABD

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EDUCATION

- 06/2027 (Exp.) **Ph.D. in Communication (STEM)**
University of California, Santa Barbara, Department of Communication
Areas of focus: health communication, persuasion, computational and experimental methods
Dissertation: *Multimodal Emotional Appeals in Anti-Vaping Videos: A Computational Approach to Fear–Hope Flow and Message Effectiveness*
Committee: Jiaying Liu (Chair), Robin Nabi, Yingdan Lu (Northwestern University)
- 12/2019 **M.A. in Communication Management**
University of Southern California, Annenberg School for Communication and Journalism
Areas of focus: marketing, management, and computer-mediated communication
- 06/2017 **B.A. Hons in International Communications Studies with German**
The University of Nottingham Ningbo China
Areas of focus: cultural studies, film and media studies, media industry
- 06/2016 **Exchange: B.A. Hons in Culture, Film and Media**
The University of Nottingham (09/2015–06/2016)

RESEARCH AREAS

Persuasion and message effectiveness · Strategic communication and health campaigns · Generative AI in persuasive messaging · Audience segmentation and message tailoring · Computational and experimental methods · Misinformation, information credibility, and AI-generated media · Human–computer interaction · Data curation and open science · Neurophysiological and psychophysiological measurement (fMRI, eye-tracking)

PUBLICATIONS

Peer-Reviewed Publications

- [1] Liu, X., Lu, Y., Peng, Q., Qian, S., Peng, Y., & Shen, C. (2026). Seeing the surreal: Mapping surrealism in photorealistic AI-generated images using large language models. *Computational Communication Research*, 8(2), 1–48. <https://doi.org/10.5117/CCR2026.2.6.LIU>
- [2] Peng, Q., Lu, Y., Peng, Y., Qian, S., Liu, X., & Shen, C. (2025). Crafting synthetic realities: Examining visual realism and misinformation potential of photorealistic AI-generated images. In *Proceedings of Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)* (pp. 1–12). ACM. <https://doi.org/10.1145/3706599.3719834>
- [3] Hong, J., Peng, Q., & Williams, D. (2020). Are you ready for artificial Mozart and Skrillex?: An experiment testing expectancy violation theory and AI music. *New Media & Society*, 23(7), 1920–1935. <https://doi.org/10.1177/1461444820925798>

Datasets

- [1] **Peng, Q.**, Lu, Y., Peng, Y., Qian, S., Liu, X., & Shen, C. (2025). *AI-generated image dataset* [Data set]. Harvard Dataverse, V1. <https://doi.org/10.7910/DVN/MGJPBL>
Publicly archived, curated corpus of photorealistic AI-generated images with human-annotated realism and content labels; supports *Crafting Synthetic Realities* (CHI EA '25)

Registered Report Accepted in Principle

- [1] Rathje, S., Asimovic, N., . . . , **Peng, Q.**, . . . , & Van Bavel, J. J. Testing the causal impact of social media reduction around the globe. *Nature*. https://doi.org/10.31234/osf.io/ujtxa_v2

Manuscripts Under Review

- [1] Gonzales, A. L., Wang, L. H., Kim, Y. W., & **Peng, Q.** Meso-level theorizing the digital divide: A model of institutional capacity for digital equity. (*Revise & resubmit at Communication Research*)
- [2] Worsdale, A., . . . , **Peng, Q.**, . . . , & Liu, J. Depressive symptoms alter the predictive value of neural responses to vaping prevention messages in young adults who vape. (*Submitted to Nicotine & Tobacco Research*)
- [3] **Peng, Q.**, Duong, H., Shi, R., & Liu, J. Contradiction disrupts and reinforcement plateaus: The interplay of news and user comments on norm perceptions in the digital age. (*Submitted to Mass Communication and Society*)
- [4] Ismail, I.*, **Peng, Q.***, Liu, J., & Oh, V. Y. Specific mixed emotional headlines drive online media engagement over and above positivity and negativity. (*Submitted to Social Science Computer Review*)
*Equal contribution
- [5] Gonzales, A. L., Kim, Y. W., Wang, L. H., & **Peng, Q.** The future of digital equity. (*Submitted to New Media & Society*)
- [6] **Peng, Q.** Mapping the discourse network of cervical cancer screening on Reddit: A topic-network approach. (*Submitted to Hawaii International Conference on System Sciences (HICSS)*)

Manuscripts in Preparation

- [1] Liu, J., **Peng, Q.**, Malik, M., Wang, Y., Norton, E., Markey, C., Ye, T., & Sweet, L. H. Identifying optimal cognitive, social, and emotional profiles of anti-vaping messages for young adult vapers: Insights from interpretable machine learning analysis. (*Manuscript ready; targeted for American Journal of Preventive Medicine*)
- [2] Ye, T., . . . , **Peng, Q.**, . . . , & Liu, J. Occipitoparietal response to vape packaging with food-cues indexes visuospatial salience and predicts future vaping frequency in young adults. (*Manuscript ready*)
- [3] **Peng, Q.** What do we measure when we measure emotion? A four-level framework for communication research. (*Manuscript ready; targeted for Annals of the International Communication Association*)
- [4] **Peng, Q.**, & Liu, J. Generative AI for scalable message tailoring: Differential persuasive effects across socioeconomic and value-based subgroups. (*Manuscript in preparation*)
- [5] Zhao, S., Yu, H., Wang, Y., **Peng, Q.**, Ye, T., Sweet, L., & Liu, J. A neural signature of vaping and smoking cues. (*Manuscript in preparation; targeted for PNAS*)

- [6] **Peng, Q.**, . . . , & Liu, J. Looking without thinking, feeling without looking: Discrete emotions drive visual attention and persuasion resistance through independent pathways in anti-smoking messages. (*Manuscript in preparation; targeted for Media Psychology*)
- [7] **Peng, Q.**, & Lu, Y. When modalities disagree: A computational framework for measuring cross-modal emotion divergence. (*Manuscript in preparation; targeted for Communication Methods and Measures*)
- [8] **Peng, Q.**, & Nabi, R. Empowering young adults against early-onset cancer: Balancing fear, hope, and psychological reactance in health messaging. (*Data collection in progress*)

Media Coverage

- 04/2026 Cha, A. E. (2026, April 9). Social media addiction? This detox could reverse a decade of damage. *The Washington Post*. [\[Link\]](#)
Coverage of Rathje et al., *Nature* registered report

CONFERENCE PRESENTATIONS

- [C.19] Ismail, I.^{*}, **Peng, Q.**[†], Liu, J., & Oh, V. Y. (2026, Nov). Pride-anger and beyond: Mixed emotion specificity in online media engagement. *NCA 112th Annual Convention (Communication and Social Cognition Division)*, New Orleans, USA.
Top Student Paper Award
*Equal contribution
- [C.18] **Peng, Q.**[†], & Liu, J. (2026, Nov). Generative AI for scalable message tailoring: Differential persuasive effects across socioeconomic and value-based subgroups. *NCA 112th Annual Convention*, New Orleans, USA.
- [C.17] **Peng, Q.**[†] (2026, Nov). Rethinking emotion in communication research: A four-level framework and computational measurement guide. *NCA 112th Annual Convention (Communication and the Future Division)*, New Orleans, USA.
Top Student Paper Award
- [C.16] **Peng, Q.**[†], Liu, M., & Liu, J. (2026, Jul). Generative AI for scalable anti-vaping message tailoring: Differential persuasive effects across socioeconomic and value-based subgroups. *Summer 2026 NIH Tobacco Regulatory Science (TRS) Meeting (Poster)*, Bethesda, MD, USA.
- [C.15] Liu, J., **Peng, Q.**[†], Malik, M., Wang, Y., Norton, E., Markey, C., Ye, T., & Sweet, L. H. (2026, Jul). Identifying optimal cognitive, social, and emotional profiles of anti-vaping messages for young adults who vape: Insights from interpretable machine learning analysis. *Summer 2026 NIH Tobacco Regulatory Science (TRS) Meeting (Oral Presentation)*, Bethesda, MD, USA.
- [C.14] Liu, J., Markey, C., Ye, T., McMains, J., Wang, Y., **Peng, Q.**, Norton, E., & Sweet, L. H. (2026, Jul). Occipitoparietal response to vape packaging with food cues predicts subsequent vaping frequency in young adults: The role of impulsive sensation seeking. *Summer 2026 NIH Tobacco Regulatory Science (TRS) Meeting (Poster)*, Bethesda, MD, USA.
- [C.13] Liu, J., **Peng, Q.**[†], Malik, M., Wang, Y., Norton, E., Markey, C., Ye, T., & Sweet, L. H. (2026, Jun). Identifying optimal cognitive, social, and emotional profiles of anti-vaping messages for young adult vapers: Insights from interpretable machine learning analysis. *76th ICA Annual Conference*, Cape Town, South Africa.
- [C.12] Liu, X., Lu, Y., **Peng, Q.**[†], Qian, S., Peng, Y., & Shen, C. (2026, Jun). Seeing the surreal: Mapping surrealism in photorealistic AI-generated images using large language models. *76th ICA Annual Conference*, Cape Town, South Africa.

- [C.11] Ye, T., Norton, E., Wang, Y., **Peng, Q.**, Sweet, L. H., & Liu, J. (2026, Jun). Social influence vs. self relevance: Neural correlates of the third-person effect in social appeal anti-vaping messages among young adult vapers. *76th ICA Annual Conference, Cape Town, South Africa*.
- [C.10] **Peng, Q.[†]**, & Liu, J. (2026, May). Generative AI for scalable message tailoring: Differential persuasive effects across socioeconomic and value-based subgroups. *Comm Horizons, Davis, USA*.
- [C.9] Liu, X., Lu, Y., **Peng, Q.**, Qian, S., Peng, Y., & Shen, C. (2026, May). Seeing the surreal: Mapping surrealism in photorealistic AI-generated images using large language models. *Comm Horizons, Davis, USA*.
- [C.8] Liu, J., **Peng, Q.[†]**, Malik, M., Wang, Y., Norton, E., Markey, C., & Sweet, L. H. (2026, Mar). Identifying optimal cognitive, social, and emotional profiles of anti-vaping messages for young adult vapers: Insights from interpretable machine learning analysis. *Annual Meeting of SRNT, Baltimore, USA*.
- [C.7] Norton, E., Markey, C., Wang, Y., **Peng, Q.**, Sweet, L. H., & Liu, J. (2026, Feb). Neurocognitive predictors of vaping reduction: Insights from Public Service Announcement exposure and machine learning. *International Neuropsychological Society (INS), Philadelphia, USA*.
- [C.6] **Peng, Q.[†]**, Duong, H., Shi, R., & Liu, J. (2024, Nov). Congruent and incongruent norms: The impact of news and user comments on norm perceptions in the digital age. *NCA (Health Communication Division), New Orleans, USA*.
- [C.5] **Peng, Q.[†]**, Lu, Y., Peng, Y., Qian, S., Liu, X., & Shen, C. (2024, Sep). A first analysis of the misinformation potential of AI-generated images: A cross-platform investigation of their content, context, and impact. *Trust and Safety Conference (Poster), Stanford University, USA*.
- [C.4] **Peng, Q.[†]**, Lu, Y., Peng, Y., Qian, S., Liu, X., & Shen, C. (2024, Jun). A first analysis of the misinformation potential of AI-generated images: A cross-platform investigation of their content, context, and impact. *74th ICA Annual Conference, Gold Coast, Australia*.
- [C.3] **Peng, Q.[†]**, Liu, M., & Williams, D. (2020, Nov). The moderating effect of player types: Who doesn't quit? *NCA (Game Studies Division), virtual*.
- [C.2] Liu, M., Lynn, M., & **Peng, Q.** (2020, Nov). Effectiveness of virtual reality interventions in reducing depressive symptoms: A meta-analysis. *NCA, virtual*.
- [C.1] Hong, J., **Peng, Q.**, & Williams, D. (2019, Jun). Are you ready for artificial Mozart and Skrillex?: An experiment testing expectancy violation theory and AI music. *ICA (Communication and Technology Division), Washington, D.C., USA*.

[†]Presenting author

HONORS, AWARDS & SCHOLARSHIPS

11/2026	Top Student Paper Award, Communication and the Future Division, NCA 112th Annual Convention Paper: <i>Rethinking Emotion in Communication Research: A Four-Level Framework and Computational Measurement Guide</i> (solo-authored)
11/2026	Top Student Paper Award, Communication and Social Cognition Division, NCA 112th Annual Convention Paper: <i>Pride-Anger and Beyond: Mixed Emotion Specificity in Online Media Engagement</i> (co-first author)
2026 Fall	Dissertation Fellowship, Department of Communication, UCSB One-quarter fellowship with full release from teaching duties to support dissertation completion

08/2026	Student Research Award, Mass Communication and Society Division, AEJMC (\$2,500) Project: <i>Generative AI for Scalable Health Message Tailoring: Differential Persuasive Effects across Socioeconomic and Value-Based Subgroups</i>
06/2026	Social Impact Award, Department of Communication, UCSB (\$2,000)
2017	Zhejiang Provincial Outstanding Graduates, Ministry of Education
2016	Zhejiang Provincial Scholarship, Ministry of Education (RMB 12,000)
2014–2016	Best Student of the Year, University of Nottingham (Top 1, two consecutive years; RMB 30,000)
2014–2016	President's Scholarship, University of Nottingham (two consecutive years; RMB 30,000)
2013–2014	Dean's Scholarship, University of Nottingham (Top 10%; RMB 6,000)

FUNDING

Internal Funding

2026	Center for AI and Society Grant, UCSB Role: Graduate Research Assistant (PI: Prof. Jiaying Liu)	\$8,000
08/2025	Social Science Research Grant Program (SSRGP), UCSB ISBER Role: Graduate Research Assistant (PI: Prof. Jiaying Liu)	\$7,600
08/2025	Individualized Professional Skills (IPS) Grant Program, UCSB Graduate Division	\$1,200
02/2025	Graduate Research Funding, Department of Communication, UCSB	\$500
06/2019	CKIDS DataFest Research Funding, USC Viterbi School of Engineering	\$3,600

External Funding

2025	TRDRP Predoctoral Fellowship Award, Tobacco-Related Disease Research Program, University of California Office of the President Project: <i>Emotion-Driven Persuasion in Anti-Vaping Campaigns: A Computational and Experimental Approach</i> Role: Principal Investigator (Graduate Student) (Shortlisted; full application submitted; not funded)	\$181,539 (proposed)
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Travel Awards & Grants

2026	Michael Haley Travel Grant, International Communication Association (ICA)	\$900
04/2026	Doctoral Student Travel Grant, Academic Senate, UCSB	\$1,500
2026	Health Communication Division Travel Award, ICA 76th Annual Conference	\$200
2024	Game Studies Division Travel Grant, ICA 74th Annual Conference, Gold Coast, Australia	\$500
2024–2026	Travel Grant, Graduate Student Association (GSA), UCSB Graduate Division (three consecutive years)	
2024–2026	Travel Funding, Department of Communication, UCSB (three consecutive years)	

RESEARCH EXPERIENCE

University of California, Santa Barbara		Santa Barbara, CA
03/2024–Present	Communication, Health, and Emerging Media (CHARM) Lab Department of Communication	

Advisor: Prof. Jiaying Liu

Research on experimental methods, neurophysiological tools (e.g., fMRI), and psychophysiological measures (e.g., eye-tracking)

Graduate Research Fellow on the following projects:

• **Project Leader on Eye-tracking & Discrete Choice Experiment**

Behavioral intervention to mitigate nicotine addiction in digital environments (R01CA289552; NIH NCI & FDA Center for Tobacco Products)

PI: Prof. Lucy Popova (Georgia State University); Co-I: Prof. Jiaying Liu (UCSB)

UCSB subaward total cost: \$357,333

• **Lead Data Analyst on Computational & AI-Assisted Analysis of Message Effects**

Neuroimaging approaches to improve prediction of smoking initiation and nicotine use escalation among young adult electronic nicotine delivery systems users (K01DA049292; NIH/NIDA & FDA Center for Tobacco Products)

PI: Prof. Jiaying Liu (UCSB); Total: \$1,305,218

• **Graduate Student Project Leader**

Scaling socioeconomic and value-based personalization in health messaging: A thin-AI approach to advancing communication equity (Health Communities Project, ISBER, UCSB)

PI: Prof. Jiaying Liu (UCSB); Total: \$7,600

Computational Multi-Modal Communication (CMMC) Lab

03/2023–Present Principal Investigators: Prof. Cuihua Shen, Prof. Yilang Peng, Prof. Yingdan Lu
Led and participated in research projects using computational methods based on Python/R

Graduate Research Fellow on the following project:

• **Project Leader on misinformation potential of photorealistic AI-generated images**

Understanding How Visual Features of Misinformation Influence Credibility Perceptions (CNS-2150716; National Science Foundation)

PIs: Prof. Cuihua Shen, Prof. Yilang Peng, Prof. Yingdan Lu

Total: \$500,000

Northwestern University

Evanston, IL

10/2024–Present **Computational Media and Politics (COMAP) Lab**

School of Communication

Principal Investigator: Prof. Yingdan Lu

University of Southern California

Los Angeles, CA

01/2019–06/2020 **Game Studies Research Group & CKIDS DataFest**

Annenberg School of Communication & Viterbi School of Engineering

Principal Investigator: Prof. Dmitri Williams

SELECTED WORKSHOPS

08/2025 **Summer Institute in Computational Social Science (SICSS-Stanford)**

Stanford University; IPS funding (\$1,200)

INVITED TALKS & GUEST LECTURES

09/2026 **Peng, Q.** Computational methods in communication research. COMM 88 *Communication Research Methods*, University of California, Santa Barbara.

06/2026 **Peng, Q.** “Don’t make it about me”: When AI-tailored health messages help, hurt, and why it depends on who you are. *Affect, Ambivalence, and Adjustment Lab*, Department

of Psychology, School of Humanities and Behavioural Sciences, Singapore University of Social Sciences.

- 04/2026 Liu, J., & **Peng, Q.** Understanding health message effectiveness: Message profiles, neural responses, and behavior change. *Health Communication Symposium, Department of Communication, UCSB.*
- 03/2026 Liu, J., & **Peng, Q.** Personalizing vaping prevention messages among young people with generative AI: What we learned. *Tobacco Prevention Summit, Santa Barbara County.*
- 01/2026 **Peng, Q.** Internet, technology & health. *COMM 155 Health Communication, University of California, Santa Barbara.*
- 04/2025 **Peng, Q.** Digital media in times of well-being. *A Symposium on Digital Media & Well-Being, Department of Communication, UCSB.*

TEACHING EXPERIENCE

Independent Instructor of Record (UCSB)

- 2025 Summer A COMM 89 Communication Theories (~50 students)
Designed syllabus, lectures, in-class activities, and assessments
- 2024 Summer B & 2025 Spring COMM 139 Communication and Emotion (~50 students)
Designed syllabus, lectures, in-class activities, and assessments; delivered in-person lectures

Teaching Assistant (UCSB)

- 2026 Summer B COMM 117 Persuasion (Prof. Robin Nabi; ~90 students)
- 2026 Spring COMM 187 Data Science (Prof. Prateekshit Pandey; ~90 students)
- 2026 Winter COMM 155 Health Communication (Prof. Jiaying Liu; ~90 students)
- 2025 Fall COMM 88 Communication Research Methods (Prof. Amy Gonzales; ~90 students)
- 2025 Summer A INT 93LS Research in STEM: Decoding Bias (Prof. Musa Malik; ~30 students)
- 2025 Winter COMM 148 Risk Communication (Prof. Katie Elder; ~90 students)
- 2024 Spring & Fall COMM 89 Communication Theories (Prof. Walid Afifi; ~90 students)
- 2024 Winter COMM 139 Communication and Emotion (Prof. Robin Nabi; ~90 students)
- 2023 Fall COMM 137 Global Communication (Prof. Michael Stohl; ~90 students)

Teaching Assistant (The University of Nottingham Ningbo China)

- 02/2017–05/2017 Industrial Marketing and Culture (Prof. Dirk Moosmeyer; ~50 students)

WORK EXPERIENCE

NorthKing Information Technology Co. LTD

Beijing, China

09/2020–09/2023 Brand & Marketing Manager

Led integrated brand strategy and corporate communications for a Beijing-based financial-IT services firm serving major banks; designed B2B campaigns spanning earned, owned, and paid media

Directed market and user-experience (UX) research; translated audience insights into messaging, positioning, and public-relations strategy

Managed client and stakeholder relationships, social media presence, and event programming, including industry conferences and executive forums

Recognized as *Outstanding Employee of the Year* for measurable contributions to brand visibility

USC Games

Los Angeles, CA

06/2018–06/2019 **Video Editor**

Produced and edited video content supporting USC Games programming and showcases

SEGA Stories International

Los Angeles, CA

05/2018–08/2018 **Branded Entertainment Intern**

Conducted brand and audience research to inform strategic communication and creative-development decisions

Supported cross-stakeholder coordination and industry events that extended SEGA's storytelling presence

Reviewed and provided coverage on creative submissions for brand and audience fit

Los Angeles Chinese Film Festival

Los Angeles, CA

04/2018–11/2018 **Marketing & PR Specialist**

Designed and executed integrated marketing and public-relations campaigns across social media, content production, and earned media

Produced editorial and interview content amplifying festival programming and partnerships

CCTV Weihai Movie & TV Base, Beijing Office

Beijing, China

08/2016–09/2016 **Intern Program Producer**

Supported pre-production for the talent show *Heroes from Youngsters*

SINA Education Channel

Beijing, China

06/2016–08/2016 **Intern Editor**

Managed social media marketing, live broadcast show planning, and online reporting

Received a 10% salary raise for outstanding performance

ACADEMIC SERVICE

Reviewer

Journals

BMC Public Health

Health Communication

Journal of Health Communication

Journal of Applied Communication Research

International Journal of Communication

Conferences

International Communication Association (Health Communication Division, Computational Methods Division)

National Communication Association

International Conference on Computational Social Science (IC²S²)

Hawaii International Conference on System Sciences (HICSS)

Mentorship

Undergraduate RAs Supervised and trained 10+ RAs in the CHARM Lab on data collection, coding, and analysis
Calista Schmidt — recipient of the Steven H. Chaffee Undergraduate Award for Excellence in Research, UCSB Department of Communication (2025)

High School Vera Lew (paper submitted to ICA 2026)
Yitong Huang
Mabelle Liu (abstract submitted to NIH Tobacco Regulatory Science Conference)

Department Service

2025 Fall Grad Buddy Mentor, Department of Communication, UCSB
09/2024 Panelist, COMM Jumpstart 2024, Department of Communication, UCSB

University Service

05/2026 Panelist, IPS Meet & Eat (Whole PhD 2026 Series), Professional Development, UCSB Graduate Division

PROFESSIONAL AFFILIATIONS

International Communication Association (ICA)
National Communication Association (NCA)
Association for Education in Journalism and Mass Communication (AEJMC)
Society for Research on Nicotine and Tobacco (SRNT)
NIH/FDA CTP-Funded Tobacco Regulatory Science (TRS) Research Community (Trainee)

SKILLS

Programming: Python, R, SPSS, Mplus, L^AT_EX

Software: Microsoft Office, Excel, Adobe Photoshop, Premiere Pro, InDesign

Languages: Chinese (Native), English (IELTS 8.5/C2), German (B2), Korean (TOPIK 3/B1)

Certificates: Methods and Statistics in Social Sciences Specialization; Adobe Certified Associate in Visual Design (Photoshop); Adobe Certified Associate in Digital Video (Premiere Pro); Adobe Certified Associate in Print & Digital Media Publication (InDesign)

August 19, 2026